

Unit 3 - Basics of Client- Server and Web Technologies

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WWW and Internet

- World Wide Web - a network of networks - first proposed by Tim Berners-Lee (1989) at CERN
- First web-client and web-server was written and executed
- Uses HTTP - HyperText Transfer Protocol

Client-Server Protocols

- **HTTP** - Hyper Text Transfer Protocol - used by browsers (clients) to request web pages from servers.
- **HTTPS** - Secure transfer of webpages
- **FTP/FTPS** - File Transfer Protocol - for transfer of files between machines
- **SMTP** - Simple Mail Transfer Protocol - For transfer of mail servers
- **POP3** - For downloading mails from a server
- **IMAP** - Internet Message Access Protocol -Management of mails across multiple clients from server
- **SSH** - Secure SHell - For remote logins

Client Server Protocols

- **DNS** - Domain Naming Services - translates domain names (like ibab.ac.in) into IP addresses.
- **DHCP** - Dynamic Host Configuration Protocol - Allocates a IP address and network configuration for systems dynamically.
- **TCP/IP** - Transmission Control Protocol/Internet Protocol - Universal protocols for Http/FTP and other communications within the Transport layer:
 - For transport layer - breaks down the data into packets for reliable / not-guaranteed communication

Network Ports

- A numerical channel (0 - 65535) which identifies which application to direct the traffic to.
- Well-known ports
 - Http - 80
 - Https - 443
 - SMTP - 25

Http

- **Methods** to communicate with server : GET/POST/PUT/DELETE/HEAD
- **Response Codes :**
 - 200 - OK,
 - 403 - Access issues ,
 - 404 - Resource not available.
 - 500 - Internal Server Error
 - 301 - Redirect
- **Headers**
 - Carry information about request/response
 - Cookies - for storing session / application data

Markup Languages

- Languages that render content
- HTML - Hyper Text Markup Language - for rendering web-pages
- XML - eXtensible Markup Language for storing, transporting generic data across systems and services.
- Latex - for rendering manuscripts

HTML

HTML

Skeleton

CSS (Cascading Stylesheet)

Flesh

JS (Javascript)

Functionality

Basics of HTML

- Paired tags/elements
- **<html>/<html>**, - encloses all the HTML content
- **<head>/</head>** - has meta information (such as charset, viewport (mobile device compatibility), links to stylesheets, javascript, inline style etc
- **<body>/</body>** - main content of HTML
- Attributes - Features of an element eg. ****
- Hyperlinks **** - relative location or absolute location

CSS

- Cascading Style Sheet
- Provides attributes such as color, background, alignment, etc to various elements of the HTML.
- Can be done :
 - As attributes of elements
 - Inline in the same html
 - Imported as a separate css

Javascript

- Client side programming language -
- for basic checks such as empty strings etc
- Executed by the client (browser) on events such as `onClick()`, `onSubmit()` etc.
- Imported in the html or coded inline in script
- Forwards client request to server upon successful validation
- Demo

Bonus : `<div>` and `<span>`

- Containers for css / js
- `<div>` - for grouping blocks of elements
- `<span>` - inline for parts of elements

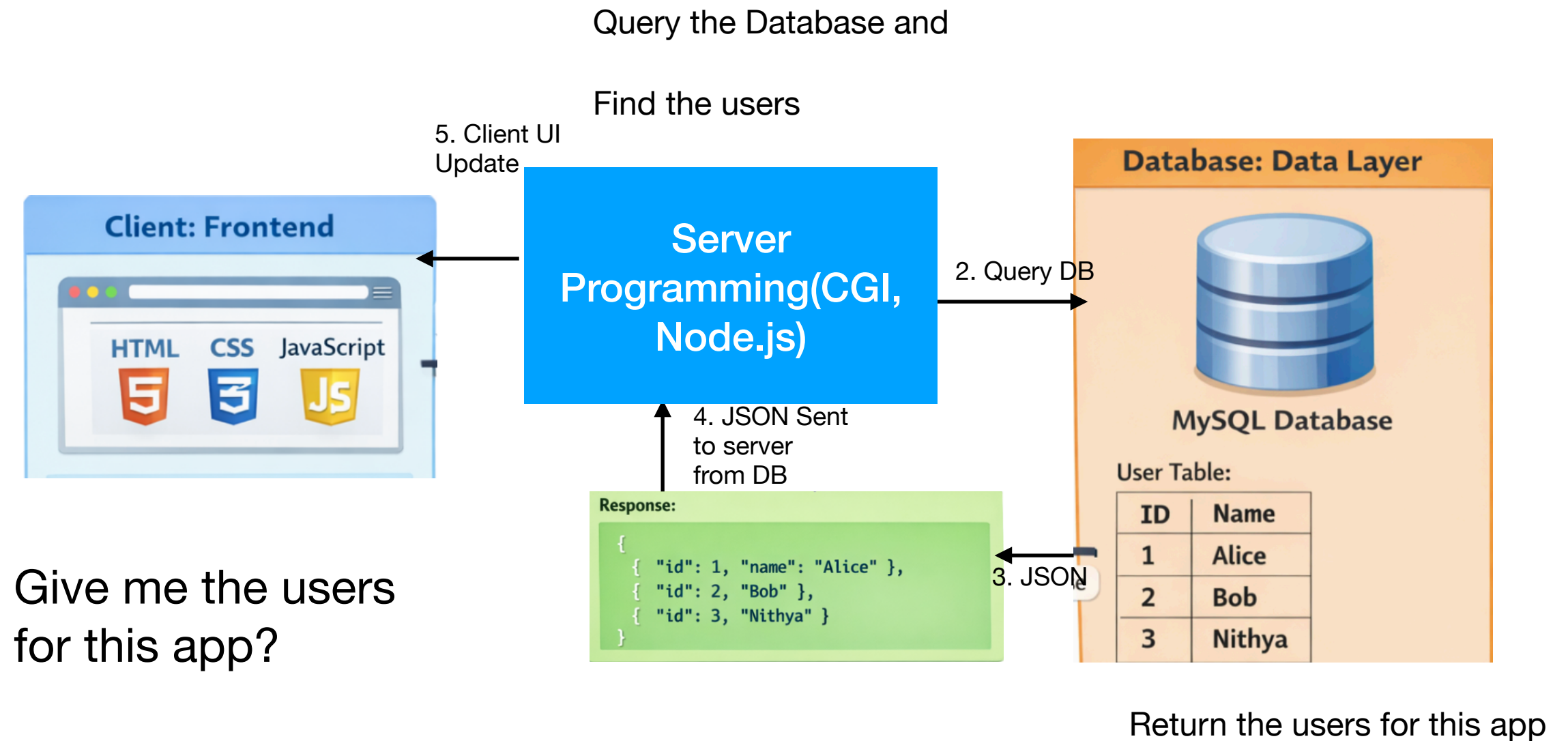
```
<div id="special">  
  <h1>About Us</h1>  
  <p>This is a paragraph about our institute.</p>  
</div>
```

```
<p>In this text, this <span id="coloured" style="color: red;">word</span> is red.</p>
```

Tutorials on HTML, CSS, JS

- Mozilla Developer Network - <https://developer.mozilla.org/en-US/docs/Web/HTML/Reference/Elements>
- W3 Schools - <https://www.w3schools.com/html/>

Full Stack Example



Agentic AI

